MATH1231 Mathematics 1B – Algebra

Section 13 - Continuous Random Variables

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Slides: David Angell, Laure Helme-Guizon, Sean Gardiner and others

We have been studying discrete random variables, where the variable takes countably many "separate" values each with its own probability. We now look at continuous random variables, in which the possible values cover a whole interval of the real line. More precisely:

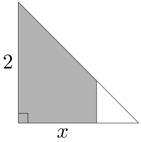
Definition (continuous random variables)

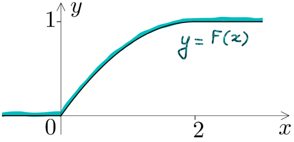
A random variable is said to be **continuous** if its cumulative distribution $F(x)$ is a continuous function.

Example 1.

a) We can see from the cumulative distribution function graphed on earlier that the geometric distribution is not a continuous random variable.

b) Choose a point at random from a right–angled isosceles triangle with side 2 as shown,





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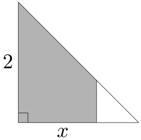

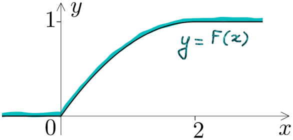

and let X be the x–coordinate of this point. It seems intuitively reasonable that $\mathbb{P}(X \leqslant x)$ should be the ratio of the shaded area to the whole area of the triangle, and a little geometry gives the second case in the formula

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x - \frac{x^2}{4} & \text{if } 0 \leqslant x \leqslant 2 \\ 1 & \text{if } x > 2. \end{cases}$$

From our knowledge of continuous functions, confirmed by the graph, we see that F is a continuous function and hence that X is a continuous random variable.

Investigation 2. For a continuous random variable X, what is the probability that X has any specific value?

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Let X be the downtime of a particular computer system, measured in minutes, on a randomly chosen day.

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Example of a random variable which is neither discrete nor continuous.
Let X be the downtime of a particular computer system, measured in minutes, on a randomly chosen day. Then the range of X is the interval $[0, 1440]$ which is not countable, so X cannot be discrete. On the other hand, there will be a non–zero probability that X has the value 0 exactly (no downtime), as well as a non–zero probability that X has the value 1440 exactly (down all day); so by the previous comment, X cannot be continuous either.

challenge : Suppose that $\mathbb{P}(X = 0) = 0.5$ and $\mathbb{P}(X = 1440) = 0.1$. Draw a possible graph for the cumulative distribution function of X. Hence confirm from the definition that X is not a continuous random variable.

In order to find means and variances of continuous random variables we shall need something analogous to the probability of obtaining a specific value of a random variable.

Definition (probability density function)

Let X be a continuous random variable with cumulative distribution function $F(x)$. The **probability density function** of X is defined by

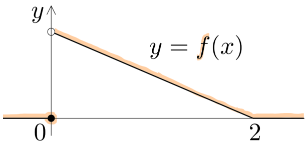
$$f(x) = F'(x) \quad \text{if } F \text{ is differentiable at } x \ ,$$

and

$$f(x) = \lim_{t \to x^-} F'(t) \quad \text{if } F \text{ is not differentiable at } x \ .$$

Example 3. For the "triangle" example, we have

$$f(x) = \begin{cases} 0 & \text{if } x \leqslant 0 \\ 1 - \dfrac{x}{2} & \text{if } 0 < x \leqslant 2 \\ 0 & \text{if } x > 2. \end{cases}$$



Note that the value of f at 0 is given by differentiating F from the left at $x = 0$.

Theorem (Properties of the probability density function.)

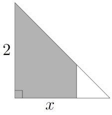
Let X be a continuous random variable with cumulative distribution function $F(x)$ and probability density function $f(x)$. Then

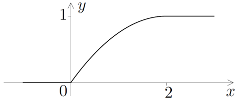
1. for all real x we have $f(x) \geqslant 0$;
2. for all real x we have $F(x) = \displaystyle\int_{-\infty}^{x} f(t)\,dt$;
3. $\displaystyle\int_{-\infty}^{\infty} f(t)\,dt = 1$;
4. if $a \leqslant b$ then $\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a < X \leqslant b) = \displaystyle\int_{a}^{b} f(x)\,dx$.

Proof (sketch).

1. By definition $f = F'$, and we know that F is increasing.
2. Definition of f, Second Fundamental Theorem of Calculus, and $\displaystyle\lim_{x\to-\infty} F(x) = 0$.
3. Property 2 and $\displaystyle\lim_{x\to\infty} F(x) = 1$.
4. Second Fundamental Theorem of Calculus, and $\mathbb{P}(a < X \leqslant b) = F(b) - F(a)$.

Back to the triangle example 4. Choose a point at random from a right–angled isosceles triangle with side 2 as shown,





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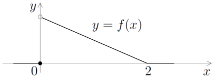

and let X be the x–coordinate of this point. Find

a) $\mathbb{P}\left(\frac{1}{2} < X < 1\right)$ b) $\mathbb{P}(X = 1)$ c) $\mathbb{P}(X < 1)$ d) $\mathbb{P}(X \geqslant 1.9)$ e) $\mathbb{P}\left(-\frac{1}{2} < X < 3\right)$

Solution.

a) $\mathbb{P}\left(\dfrac{1}{2} < X < 1\right) =$

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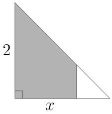

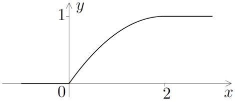

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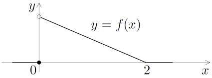

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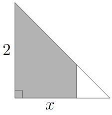

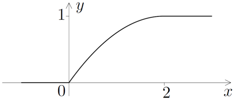

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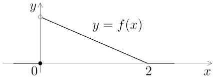

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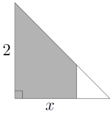

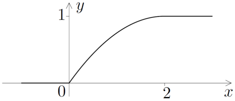

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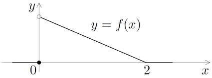

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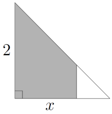

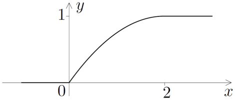

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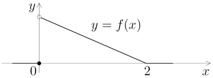

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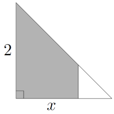

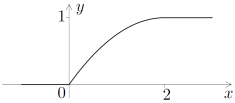

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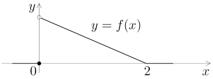

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Investigation 5. What is the meaning of $f(x)$, where f is the probability density function and x is some real number?

- Is $f(x)$ is the probability that $X = x$?
- If not, can we nevertheless attach some meaning to it?

Maybe looking at an example can help you answer the above question.

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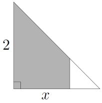

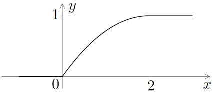

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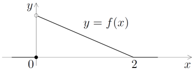

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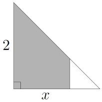

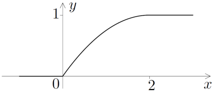

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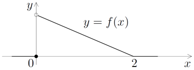

Note that $f(x)$ is NOT the probability that $X = x$, since, as we know, this probability is zero. However, the size of $f(x)$ can be considered intuitively as giving a qualitative indication of the likelihood that X is "near" x.

In the triangle example, $f(x)$ is larger for x near 0 than it is for x near 2: this indicates that a randomly chosen point in the triangle is more likely to be near the left–hand side than the right–hand tip; which is intuitively plausible.

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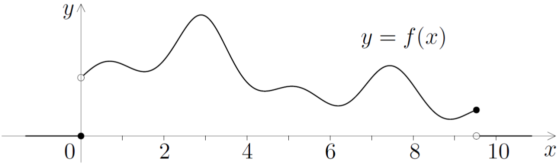
For another example, let X be the distance (in km) from the bus stop of my bus this morning. An entirely hypothetical probability density function might be as shown. Describe a situation which could result in such a graph.



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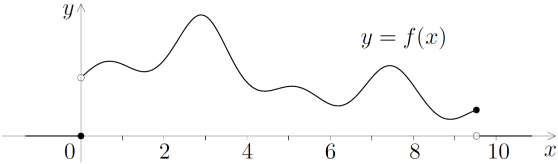

Possible answer. We can see from the graph that there is a relatively high probability of the bus being somewhere around 3 km from the stop; one might guess that at about this distance, the bus was going slowly owing to heavy traffic. Something similar happened between 7 and 8 km from the stop. The bus was never more than about 9.5km away, so perhaps this was the total length of the trip.

Example 6. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \begin{cases} c(1-x^2) & \text{if } 0 < x \leqslant 1 \\ 0 & \text{otherwise.} \end{cases}$

a) Find a positive constant c such that f is a probability density function

b) Find the corresponding cumulative distribution function.

Solution. It is usually helpful to draw a graph.

a) (i) Since $c > 0$, it is clear that $f(x) \geqslant 0$ for all x. (ii) We have

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 c(1-x^2)\,dx = \frac{2}{3}\ ,$$

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a) (i) Since $c > 0$, it is clear that $f(x) \geqslant 0$ for all x. (ii) We have

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 c(1-x^2)\,dx = \frac{2}{3}\ ,$$

and this must equal 1, so $c = \frac{3}{2}$.

Example 6. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \begin{cases} c(1-x^2) & \text{if } 0 < x \leqslant 1 \\ 0 & \text{otherwise.} \end{cases}$

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- if $x \leqslant 0$ then $F(x) = 0$,
- If $0 < x \leqslant 1$ then

$$F(x) = \int_0^x \frac{3}{2}(1-t^2)\,dt = \frac{3}{2}\,x - \frac{1}{2}\,x^3\ ,$$

- while if $x > 1$ then

$$F(x) = \int_0^1 \frac{3}{2}(1-t^2)\,dt = 1\ .$$

The last result is clearly correct since in this example, if $x > 1$ then it is *certain* that $X \leqslant x$.

Example 6, continued.

Therefore the cumulative distribution function is

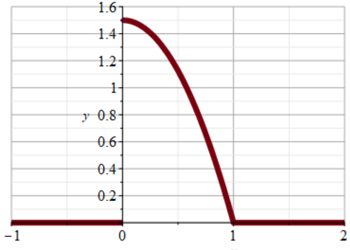
$$F(x) = \begin{cases} 0 & \text{if } x \leqslant 0 \\ \frac{3}{2}x - \frac{1}{2}x^3\ , & \text{if } 0 < x \leqslant 1 \\ 1 & \text{if } x > 1 \end{cases}$$

We should **check** that this function is continuous.

```
> f := x -> piecewise(x < 0, 0, x < 1,
  (3/2)*(1-x^2), 0);
```

$$f := x \mapsto \begin{cases} 0 & x < 0 \\ \frac{3}{2} - \frac{3 \cdot x^2}{2} & x < 1 \\ 0 & \textit{otherwise} \end{cases}$$

```
> plot(f(x), x = -1..2, discont = true,
  gridlines, y = -.02..1.6);
```




```
> F := x -> piecewise(x < 0, 0, x < 1,
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```

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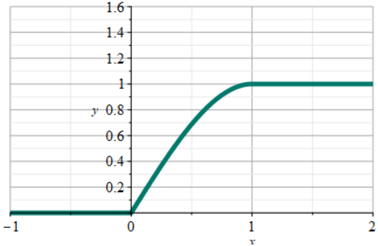

Definition (expected value of a continuous random variable)

The **mean** or **expected value** of a continuous random variable X with probability density function f is

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*What is your guess for the expected value of **a function of** a continuous random variable?*

$$\mathbb{E}(g(X)) = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\ ?$$

Theorem (Expected value of a function of a random variable)

Let X be a continuous random variable with probability density function f, and let g be a function from $\mathbb{R}$ to $\mathbb{R}$. Then the random variable $Y = g(X)$ has mean

$$\mathbb{E}(Y) = \mathbb{E}\left(g(X)\right) = \int_{-\infty}^{\infty} g(x)f(x)\,dx\ .$$

Proof. Difficult, omitted.

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Can you guess a ***different*** *formula for the variance of a continuous random variable?*

$\mathbb{V}ar(X) =$ ______________________________

Now that we know how to calculate the expected value of a function of a random variable, we can define variance and standard deviation for continuous random variables exactly as we did for discrete random variables.

Definition (variance of a continuous random variable)

Let X be a continuous random variable with mean μ. The **variance** of X is

$$\mathbb{V}\mathrm{ar}(X) = E((X - \mu)^2) \ .$$

The **standard deviation** of X, denoted $\mathrm{SD}(X)$, is the (positive) square root of the variance.

Theorem (Alternative formula for the variance)

If X is a continuous random variable, then

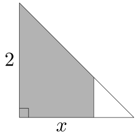
$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \ .$$

Same mnemonic: "The mean of the square minus the square of the mean".

Proof. Exercise: essentially as for discrete random variables, but with integrals instead of sums.

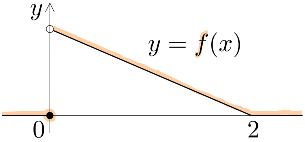
Example 7: The "triangle" again.

Choose a point at random from a right–angled isosceles triangle with side 2 as shown, and let X be the x–coordinate of this point.
Find $\mathbb{V}\mathrm{ar}(X)$ and $SD(X)$.



Solution. Recall that the probability density function of X is

$$f(x) = \begin{cases} 0 & \text{if } x \leqslant 0 \\ 1 - \dfrac{x}{2} & \text{if } 0 < x \leqslant 2 \\ 0 & \text{if } x > 2. \end{cases}$$



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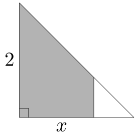

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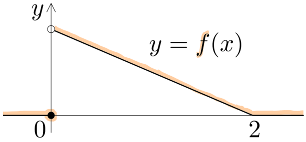

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)\,dx = \int_0^2 \left(x - \frac{x^2}{2}\right) dx = \frac{2}{3}$$

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^2 \left(x^2 - \frac{x^3}{2}\right) dx = \frac{2}{3}$$

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{2}{9} \quad \text{and} \quad \mathrm{SD}(X) = \frac{\sqrt{2}}{3} \approx 0.47\ .$$

Theorem (Scaling and shifting a continuous random variable)

Let X be a continuous random variable, and let a and b be real constants. Then the mean, variance and standard deviation of the random variable $aX + b$ are given by

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b \ ,$$

$$\mathbb{V}\mathrm{ar}(aX + b) = a^2\mathbb{V}\mathrm{ar}(X) \quad \text{and} \quad \mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X) \ .$$

Proof. If X has probability density function $f(x)$ then

$$\mathbb{E}(aX + b) = \int_{-\infty}^{\infty} (ax + b) f(x)\, dx = a \int_{-\infty}^{\infty} x f(x)\, dx + b \int_{-\infty}^{\infty} f(x)\, dx = a\mathbb{E}(X) + b \ ,$$

and the proof of the rest is identical with that for the discrete case.

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Example 8. Let X be a continuous random variable with mean equal to 12 and standard deviation equal to 3.

a) Let $Y = \frac{X+6}{3}$. Find $\mathbb{E}(Y)$ and $\mathrm{SD}(Y)$.

b) By scaling and shifting X, create a new random variable Z with mean equal to 0 and standard deviation equal to 1.

Theorem (Standardisation)

Let X be a discrete or continuous random variable with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\mathrm{ar}(X) = \sigma^2$. The random variable

$$Z = \frac{X - \mu}{\sigma}$$

has mean $\mathbb{E}(Z) = 0$ and variance $\mathbb{V}\mathrm{ar}(Z) = 1$.

Definition (Standardisation)

If $\mathbb{E}(X) = \mu$ and $\mathbb{V}\mathrm{ar}(X) = \sigma^2$, then the random variable

$$Z = \frac{X - \mu}{\sigma}$$

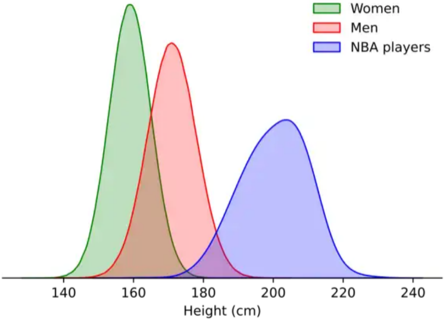
is called the **standardised** random variable obtained from X.

And now, the star among continuous random variables.

The Normal distribution

Many things closely follow a Normal Distribution:

- heights of people
- size of things produced by machines
- errors in measurements
- blood pressure
- marks on a test



Source: `https://distributionofthings.com/human-height/`

Proposition

Let μ and σ be real constants with $\sigma > 0$. Then the function

$$\phi : \mathbb{R} \to \mathbb{R} \quad \text{where} \quad \phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

is a probability density function.

Proof. (partial). It is clear that $\phi(x) \geqslant 0$ for all x, and substituting $u = (x-\mu)/\sigma\sqrt{2}$ gives

$$\int_{-\infty}^{\infty} \phi(x)\, dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2} \sigma\sqrt{2}\, du = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2}\, du \; .$$

However it can be proved (not easy!!) that

$$\int_{-\infty}^{\infty} e^{-u^2}\, du = \sqrt{\pi} \; ,$$

and thus $\int_{-\infty}^{\infty} \phi(x)\, dx = 1$. So ϕ is a probability density function.

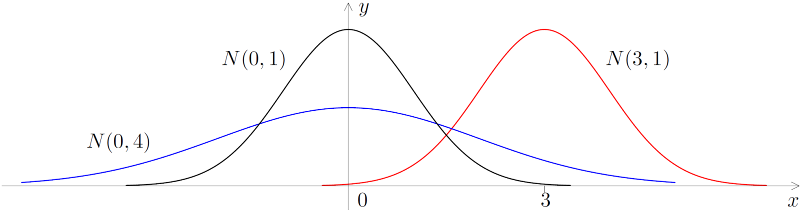
Definition (**normal distribution**)

A continuous random variable with probability density function

$$\phi : \mathbb{R} \to \mathbb{R} \quad \text{where} \quad \phi(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/2\sigma^2}$$

is said to have the **normal** or **Gaussian** distribution with mean μ and variance σ^2, and we write $X \sim N(\mu, \sigma^2)$.

The normal distribution with $\mu = 0$ and $\sigma = 1$ is called the **standard normal distribution** and is usually denoted Z, and we write $Z \sim N(0, 1)$.



Note that the normal distribution is symmetric about its mean.

The notations for the parameter of a a Normal distribution were well chosen.

We need to confirm something we stated without justification on the previous page...

Theorem (...)

If the random variable X is normally distributed, $X \sim N(\mu, \sigma^2)$, then

$$\mathbb{E}(X) = \mu \quad \text{and} \quad \mathbb{V}\mathrm{ar}(X) = \sigma^2 \ .$$

Proof. (sketch). The mean of X is given by an integral which we can split into two bits,

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} (x-\mu)\, \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/2\sigma^2}\, dx + \mu \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/2\sigma^2}\, dx \ ;$$

it is easy to show that the first integral is 0, and we already know that the second is 1. The variance

$$\mathbb{V}\mathrm{ar}(X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x-\mu) \left[(x-\mu) e^{-(x-\mu)^2/2\sigma^2} \right] dx \ ,$$

can be evaluated using integration by parts.

Exercise. Complete these calculations.

Standardising normal variables

Theorem (Standardising normal random variables)

If $X \sim N(\mu, \sigma^2)$, then $\dfrac{X-\mu}{\sigma} \sim N(0,1)$.

Conversely, if $Z \sim N(0,1)$, then $X = \sigma Z + \mu \sim N(\mu, \sigma^2)$

We already knew which mean and variance we would get. What this tells us is that after scaling and shifting a normal RV, you obtain a RV which is also normal.

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Proof. If $X \sim N(\mu, \sigma^2)$, then its probability density function is

$$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/2\sigma^2} .$$

Therefore $Z = \dfrac{X-\mu}{\sigma}$ has cumulative distribution function

$$\mathbb{P}(Z \leqslant z) = \mathbb{P}(X \leqslant \sigma z + \mu) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\sigma z + \mu} e^{-(x-\mu)^2/2\sigma^2}\, dx \;;$$

by the fundamental theorem of calculus, Z has probability density function

$$\frac{d}{dz}\mathbb{P}(Z \leqslant z) = \frac{1}{\sigma\sqrt{2\pi}}\, \sigma\; e^{-(\sigma z)^2/2\sigma^2} = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2} ,$$

and so by definition $Z \sim N(0,1)$.

Comment. Note that we have found the probability density function of Z by first considering the cumulative distribution function. This is often a good thing to do since the latter is by definition a probability, which can be grasped intuitively, whereas the probability density function has no "direct" physical meaning.

Use of the normal distribution. The normal distribution is important for various reasons.

- Many random variables in practice are found to be (at least approximately) normally distributed.
- The central limit theorem demonstrates that the normal distribution is connected in a very fundamental way with the process of averaging, for example, when we seek to measure a quantity accurately by taking the average of several separate readings – more on this in second year.
- And the normal distribution can be used to approximate other distributions, particularly the binomial, when the number of trials becomes large.

To calculate normal probabilities when $X \sim N(\mu, \sigma^2)$,
◇ the "obvious" method is to integrate the density function,

$$\mathbb{P}\left(X \leqslant x\right) \;=\; \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-(x-\mu)^2/2\sigma^2}\, dx \;.$$

Unfortunately, this cannot be done in terms of elementary functions.

◇ We can reduce to the standard normal distribution by writing

$$\mathbb{P}\left(X \leqslant x\right) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \leqslant \frac{X-\mu}{\sigma}\right) = \mathbb{P}\left(Z \leqslant \frac{X-\mu}{\sigma}\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{(x-\mu)/\sigma} e^{-z^2/2}\, dz \;,$$

but this is still impossible in elementary terms.

To get any further we have to approximate the integral by using Riemann sums (or more sophisticated techniques).

Two solutions:
◇ Values of the standard normal distribution have been tabulated, so that we can make the transformation in the previous equation and then just look up the result (Keep in mind that that the normal distribution is symmetric about its mean, it often helps).
◇ ... or ask Maple!

Standard normal probabilities $P(Z \leqslant z)$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141

Example 9.

$Z \sim N(0, 1)$.

a) $\mathbb{P}(Z < -0.81)$

$=$ ________

b) $\mathbb{P}(Z > 0.81)$

$=$ ________

c) $\mathbb{P}(|Z| < 1.96)$

$=$ ________

***Source*: course notes**

Example 9, more writing space.

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What if the random variable is normal but NOT the standard one?

Example 10.

Let $X \sim N(13, 49)$. In other words, X follows a normal distribution with mean $\mu = 13$ and variance $\sigma^2 = 49$.
Find the folowing probabilities.

a) $\mathbb{P}(X < 8) =$ ________

b) $\mathbb{P}(X > 0) =$ ________

Problem

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Example 10.

Let $X \sim N(13, 49)$. In other words, X follows a normal distribution with mean $\mu = 13$ and variance $\sigma^2 = 49$.
Find the folowing probabilities.

a) $\mathbb{P}(X < 8) =$ ________

b) $\mathbb{P}(X > 0) =$ ________

c) $\mathbb{P}(20 < X < 30) =$ ________

How would we do this using Maple? (step-by-step)

A 21st century solution

```
> with(Statistics): # Loads the statistics package. NEEDED
  mu := 13;
  sigma := 7; # In Maple you enter the SD, not the variance!
  X := RandomVariable(Normal(mu, sigma));
  Z := RandomVariable(Normal(0, 1)):
```

$$\mu := 13$$
$$\sigma := 7$$
$$X := \_R \tag{1}$$

```
> Probability(X < 8); # an unhelpful and somewhat mesmerising exact value
```

$$\frac{1}{2} - \frac{\operatorname{erf}\left(\frac{5\sqrt{2}}{14}\right)}{2} \tag{2}$$

```
> evalf(%);   # That's better!
```

$$0.2375252621 \tag{3}$$

```
> Probability(X < 8.0);  # The decimal point is another way to make Maple write the answer in
  decimal format
```

$$0.237525262026976 \tag{4}$$

```
> Probability(X > 0.0);
```

$$0.968354583883327 \tag{5}$$

```
> Probability(20 < X < 30);
Error, `<` unexpected
> # Maple is unhappy; We need to break the double inequality into 2 single inequalities.
> Probability(X < 30.0); Probability(X < 20.0 ); Probability(X < 30.0) - Probability(X < 20.0 )
```

$$0.992420780561280$$
$$0.841344746068543$$
$$0.151076034492737 \tag{6}$$

```
> Probability({X < 30.0, X > 20.0}); # The braces indicate a set:{condition 1, condition 2}
```

$$0.151076034492737 \tag{7}$$

SUMMARY. Normal distributions and Maple (and exams)

Syntax advice:

The Maple syntax to calculate the **binomial** coefficient $\binom{n}{k}$ (read '*n choose k*') is

```
binomial(n,k);
```

Maple syntax for normal distributions : Assume X follows a normal distribution with mean **3** and standard deviation **5**.

```
> with(Statistics):
  X := RandomVariable(Normal(3,5));
  # To calculate a probability
  area := evalf( Probability(X < 2) );
```

$area := 0.4207402906$

```
> # Conversly, to calculate q from the probability P(X < q):
  Quantile(X, area);
```

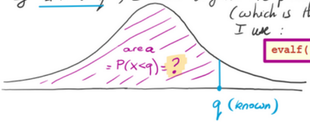
2.00000000049998

This syntax advice box is provided in the exam questions which usse Normal distributions (but not the sketches below)

Visually :

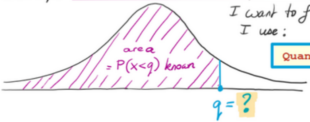
If I know q, I want to find the probability to the left of q (which is the shaded area), I use :

```
evalf( Probability(X < q) );
```



Conversly : I know the area (= the probability to the left of q) I want to find the corresponding q I use :

```
Quantile(X, area);
```



Example 11. Let T be the time in minutes taken for the light rail trip from Central Station to UNSW. Suppose that $T \sim N(16, 9)$.

a) Find the proportion of trips which take less than 15 minutes.

b) How much time should one allow in order to be 99% confident of arriving on time?

c) (next slide)

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$$\mathbb{P}(T < 15) \approx 0.3707. \quad \text{(Thanks Maple!)}$$

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....*continued*

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> restart;
> with(Statistics): # Loads the Statistics package. NEEDED
  mu := 16;
  sigma := 3; # In Maple you enter the SD, not the variance!
                                μ := 16
                                σ := 3
> Quantile(Normal(mu, sigma), 0.99); # The area to the left of the quantile is 0.99.
                            22.9790436221222
> # or directly
  Quantile(Normal(16, 3), 0.99);
                            22.9790436221222
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'Inverse probability" calculations can be performed in `Maple` by loading the `Statistics` package and using the `Quantile` command.

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$$\sigma := 3$$

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```

$$22.9790436221222$$

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> # or directly
  Quantile(Normal(16, 3), 0.99);
```

$$22.9790436221222$$

The solution to the equations $\mathbb{P}(T < t) = 0.99$ is $t = 22.99$, therefore we should allow 23 minutes in order to be 99% confident of arriving on time.

Example 11, continued. Let T be the time in minutes taken for the light rail trip from Central Station to UNSW. Suppose that $T \sim N(16, 9)$.

c) How many times this term would you expect to get from Central Station to UNSW in under 15 minutes assuming you come to campus 50 times this term?

Example 11, continued. Let T be the time in minutes taken for the light rail trip from Central Station to UNSW. Suppose that $T \sim N(16, 9)$.

c) How many times this term would you expect to get from Central Station to UNSW in under 15 minutes assuming you come to campus 50 times this term?

Solution.

c) Let X be the number of days out of the 50 days that you come in to uni this term on which the light rail trip takes less than 15 minutes. We can regard a trip as a Bernoulli trial, where a sub–15–minute trip is viewed as a "success". It seems reasonable to assume that trips on different days will be independent; so X has a binomial distribution,

$$X \sim B(50,\ 0.3707)\ .$$

The expected number of successes is

$$\mathbb{E}(X) = np = 50 \times 0.3707 = 18.54\ .$$

So we expect to arrive in under 15 minutes about 18 or 19 times in the term.

....continued

Example 12.

A food company produces *ready-to-eat salads*. Let X be the *shelf life* (in days) of a randomly chosen such salad, stored at the recommended temperature.

- From past studies, it is known that X follows a normal distribution with mean $\mu = 10$ days.
- The standard deviation σ of X depends small temperature deviations and handling conditions. The company can therefore adjust their procedures to decide on the standard deviation.

Which value(s) of the standard deviation would ensure that at least 95%of the salads are still good 7 days after being put on the shelf?

Answer: This requirement can be written as the following inequation, where σ is the unknown :

In some cases we need to standardise (even if you use Maple)

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$\mathbb{P}(X > 7) = \mathbb{P}\left(\frac{X-10}{\sigma} > \frac{7-10}{\sigma}\right) = \mathbb{P}\left(Z > \frac{7-10}{\sigma}\right) = \mathbb{P}\left(Z > -\frac{3}{\sigma}\right) = 0.95$

where $Z \sim \mathcal{N}(0,1)$.

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This is equivalent to $\mathbb{P}\left(Z < -\frac{3}{\sigma}\right) = 0.05$,

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This is equivalent to $\mathbb{P}\left(Z < -\frac{3}{\sigma}\right) = 0.05$, which happens when $\sigma = \sigma_0 \approx 1.824$.

```
> with(Statistics):
  Z := RandomVariable(Normal(0,1)):
  q := Quantile(Z, 0.05);
                              q := -1.64485362695213
> sigma_0 := -3/q;
                              sigma_0 := 1.82387049573457
```

Summary : We have shown that $\mathbb{P}(X > 7) = 0.95$ when $\sigma = \sigma_0 \approx 1.824$.

Step 2 : The requirement $\mathbb{P}(X > 7) > 0.95$
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Step 2 : The requirement $\mathbb{P}(X > 7) > 0.95$
is satisfied when $\sigma < \sigma_0 \approx 1.824$

Conclusion: At least 95% of the salads are still good 7 days after being put on the shelf provided the standard deviation σ of the salads shelf life is less than $\sigma_0 \approx 1.824$ days.

If n is large, calculating binomial probabilities can be quite tedious. For instance...

Example 13. A die is tossed 200 times. Find the probability that 30 or more sixes are obtained.

Problem

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Problem *If n is large, calculating binomial probabilities can be quite tedious. For instance...*

Example 13. A die is tossed 200 times. Find the probability that 30 or more sixes are obtained. **Solution.** To find the probability is "in principle" a fairly simple

application of the binomial distribution. If X is the number of sixes obtained from 200 rolls of the die, then $X \sim B(200, \frac{1}{6})$ and we have

$$\mathbb{P}(X \geqslant 30) \;=\; 1 - \mathbb{P}(X \leqslant 29) \;=\; 1 - \sum_{k=0}^{29} \binom{200}{k} \left(\frac{1}{6}\right)^{k} \left(\frac{5}{6}\right)^{200-k} .$$

However, there are serious difficulties in actually calculating this probability (though less so with computer assistance). For instance,

$$\binom{200}{29} = \frac{200 \times 199 \times \cdots \times 172}{29 \times 28 \times \cdots \times 1}$$

is a number of 35 digits. Instead of struggling with the calculation, . . .

....continued

... an alternative is to approximate the binomial distribution by a normal distribution Y with the same mean and variance.

Since

$$\mathbb{E}(X) = np = \frac{100}{3} \quad \text{and} \quad \mathbb{V}\mathrm{ar}(X) = npq = \frac{250}{9}$$

we shall use the approximation

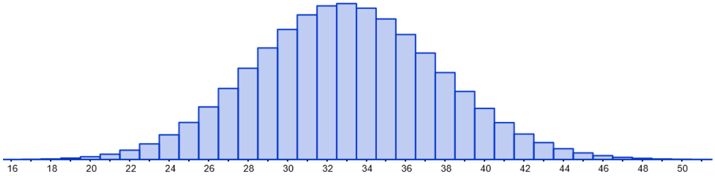
$$Y \sim N\left(\frac{100}{3}, \frac{250}{9}\right) .$$

Problem

Note that X is a discrete distribution taking integer values, while Y can take arbitrary real values. Hmmm....

How can we make this work?? What value seems to work best?

$$\mathbb{P}\,(X \geqslant 30) \;\approx\; \mathbb{P}\,(Y \geqslant \underline{\hspace{3em}}\,)$$



Recall that since X follows a binomial distribution with $\mathbb{E}(X) = np = \frac{100}{3}$ and $\mathbb{V}\mathrm{ar}(X) = npq = \frac{250}{9}$, we approximate X by $\quad Y \sim N\left(\frac{100}{3}, \frac{250}{9}\right)$.

$$\begin{aligned}
\mathbb{P}(X \geqslant 30) &\approx \mathbb{P}(Y > 29.5) && \text{(continuity correction)} \\
&= \mathbb{P}\left(\frac{Y - \frac{100}{3}}{\frac{5}{3}\sqrt{10}} > \frac{29.5 - \frac{100}{3}}{\frac{5}{3}\sqrt{10}}\right) && \text{(standardisation)} \\
&= \mathbb{P}(Z > -0.73) \;=\; 0.7673\,. && \text{(table or Maple)}
\end{aligned}$$

In fact, it is possible, using `Maple` or otherwise, to compute the exact binomial probability

$$\mathbb{P}(X \geqslant 30) = 0.7634$$

for this problem; which shows that the normal approximation (with the continuity correction) has given a satisfactory result with very little effort.

Example 14. A standard treatment for cancer is known to cure 62% of patients. A new treatment is given to 171 patients and is observed to cure 122 of them. Does this provide evidence that the new treatment is better than the standard treatment?

Solution.

Example 14. A standard treatment for cancer is known to cure 62% of patients. A new treatment is given to 171 patients and is observed to cure 122 of them. Does this provide evidence that the new treatment is better than the standard treatment?

Solution. The proportion cured under the new treatment is $\frac{122}{171} = 71\%$; so we have no reason to believe that the older treatment is superior.

We run a Hypothesis test, which is to Stats what a proof by contradiction is to maths.

H_0; Suppose that there is no difference between the two treatments; then the number of patients cured by the new treatment would be a random variable $X \sim B(171, 0.62)$. The probability of obtaining results as good as, or better than, those observed if H_0 is true would be

$$\mathbb{P}\,(X \geqslant 122)\ .$$

We use the normal approximation

$$Y \sim N(np,\, npq) = N(106.02,\, 40.2876)$$

to estimate

$$\mathbb{P}\,(X \geqslant 122) \;\approx\; \mathbb{P}\,(Y \geqslant 121.5) = \mathbb{P}\left(\frac{Y - 106.02}{\sqrt{40.2876}} \geqslant \frac{121.5 - 106.02}{\sqrt{40.2876}}\right)$$
$$= \mathbb{P}\,(Z \geqslant 2.44) = 0.0073 < 1\%\ .$$

Since this probability is very small (less than 5%), we conclude that, according to the available evidence, the new treatment is superior to the standard treatment.

This gives an outline of the use of statistical methods to answer the sort of question posed at the start of this chapter. The new treatment has cured 71% of the patients and has therefore performed better than the standard treatment. We have to decide whether or not this is just a matter of chance.

To do this we assume that there is no difference between the two treatments, and calculate the probability, under this assumption, of the result which was actually observed. Note that there is nothing particularly surprising in the fact that 122 people were cured – things would have been much the same if it had been 121 or 123. The significant fact is that 122 *or more* were cured. Therefore the relevant probability to calculate is not $\mathbb{P}(X = 122)$ but $\mathbb{P}(X \geqslant 122)$.

The result of our probability calculation shows that *if the new treatment is only as good as the old* then something extremely unlikely has occurred. Therefore it seems reasonable to rule out this possibility, and conclude that in actual fact the new treatment is better than the previous one. We might therefore recommend that the old methods be replaced by the new.

You will learn more about this kind of procedure if you do a statistical course dealing with **hypothesis testing**.

Example 15. A cereal manufacturer claims that at most 1% of its "1kg" boxes of cereal are underweight (that is, less than 1kg). When 47 randomly chosen boxes are inspected it is found that 3 of them are underweight. Is this evidence sufficient to refute the manufacturer's claim?

Solution. Assuming the manufacturer's claim is true, the number of underweight boxes is $X \sim B(47, 0.01)$. Using the normal approximation $Y \sim N(0.47, (0.68)^2)$ gives

$$\mathbb{P}(X \geqslant 3) \approx \mathbb{P}(Y \geqslant 2.5) = \mathbb{P}\left(\frac{Y - 0.47}{0.68} \geqslant \frac{2.5 - 0.47}{0.68}\right) = \mathbb{P}(Z \geqslant 2.98) = 0.0014\,.$$

Therefore the manufacturer's claim should be rejected.

Example 16. A cereal manufacturer claims that at most 1% of its "1kg" boxes of cereal are underweight (that is, less than 1kg). When 47 randomly chosen boxes are inspected it is found that 3 of them are underweight. Is this evidence sufficient to refute the manufacturer's claim?

Solution. Assuming the manufacturer's claim is true, the number of underweight boxes is $X \sim B(47, 0.01)$. Using the normal approximation $Y \sim N(0.47, (0.68)^2)$ gives

$$\mathbb{P}(X \geqslant 3) \approx \mathbb{P}(Y \geqslant 2.5) = \mathbb{P}\left(\frac{Y - 0.47}{0.68} \geqslant \frac{2.5 - 0.47}{0.68}\right) = \mathbb{P}(Z \geqslant 2.98) = 0.0014\,.$$

Therefore the manufacturer's claim should be rejected.

Alternative solution. For this problem it is not too hard to evaluate the exact binomial probability:

$$\begin{aligned}\mathbb{P}(X \geqslant 3) &= 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) - \mathbb{P}(X = 2)\\ &= 1 - \binom{47}{0}(0.01)^0(0.99)^{47} - \binom{47}{1}(0.01)^1(0.99)^{46} - \binom{47}{2}(0.01)^2(0.99)^{45}\\ &= 0.0117\,.\end{aligned}$$

What do you notice?

Note that in this case the normal approximation to the binomial distribution has failed badly! The true answer is more than 8 times our approximate calculation.

Why did this happen?

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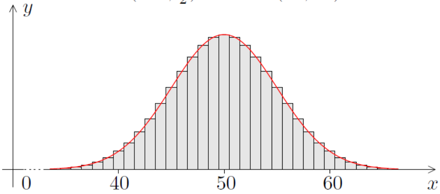
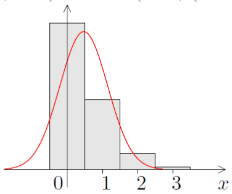
Why did this happen? One reason can be seen by looking at the following graphs. In the first we have a binomial distribution with $p = 0.5$, giving a symmetric histogram which is well approximated by the normal curve. In the second, with $p = 0.01$, the binomial distribution is highly asymmetric, and so we can scarcely expect it to be well matched by the symmetric normal curve.

$B(100, \frac{1}{2})$ and $N(50, 25)$



$B(47, 0{\cdot}01)$ and $N(0{\cdot}47, (0{\cdot}68)^2)$



Conclusion. *Approximation methods should always be used with caution!* As we are not even attempting to calculate the true answer, it should be clear that there is a lot of room for things to go badly wrong. Again, this is something you may learn more about in future statistics courses.

The end!

Thanks guys for a fun term!